CPRG 213 - Web Development 1

Description
This course introduces the fundamental concepts and technologies required for website development, beginning with examining how websites function by creating a static webpage using standard elements and logical document structure. Emphasis is placed on style rules and standards and their application on a webpage. Additional exposure to the use of advanced page scripting for the inclusion of interactivity on a website will be reinforced through hands-on website deployment.

3 Credits

Time Guidelines
The standard instructional time for this course is 75 hours.

Course Assessment

Assignments	60%
Labs	40%
Total:	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Other Course Information
Learner Engagement:

In order to be successful, the learner is expected to be engaged in learning activities for a total of 9 to 12 learning hours per course per week, which includes both in-class and out-of-class time.

Course Learning Outcomes
1. Describe how websites work.

Objectives:

1.1 Understand the role of a web server.

1.2 Describe how clients connect to web servers.

1.3 Differentiate between front-end web development and back-end web development.

1.4 Differentiate between Markup, Style Rules and Scripting languages.

1.5 Describe web development standards.

2. Create a static webpage using standard elements.

Objectives:

2.1 Describe Markup.

2.2 Define Current Markup elements and their attributes.

2.3 Distinguish between various elements.

2.4 Ensure a webpage meets current World Wide Web Consortium (W3C) validation requirements.

2.5 Explain the purpose of sectioning elements.

2.6 Describe the document outline model (DOM).

3. Apply Style rules to style a webpage.

Objectives:

3.1 Relate style rules to web development.

3.2 Describe the different ways to link style rules to a webpage.

3.3 Describe the style rules syntax.

3.4 List basic style selectors.

3.5 Explain how to use style rules for page layout.

3.6 Describe various layout models.

3.7 Differentiate and understand the relationship between elements and layout models.

3.8 Apply the models to a webpage.

4. Integrate interactive elements in a webpage.

Objectives:

4.1 Define the different types of interactive elements available in markup.

4.2 Describe the purpose of each type of interactive element.

4.3 Use interactive elements in a webpage.

5. Outline the functionalities of page level scripting languages.

Objectives:

5.1 Relate scripting languages to web development.

5.2 Describe how scripting languages impacts markup and style rules.

5.3 Define events in scripting languages.

5.4 Use variables to hold data.

5.5 Construct scripting expressions.

5.6 Demonstrate how to use logic in scripting languages.

5.7 Construct methods in scripting languages

6. Deploy a website to the internet.

Objectives:

6.1 Outline website deployment tools.

6.2 Describe the domain name system (DNS).

SAIT Policies and Procedures:
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to those in the Academic Student, Conduct, and Health, Safety, and Environment domains.

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